The Chinese University of Hong Kong, Shenzhen

Course Outline

1. Course identity

A. Course as listed at CUHK-Shenzhen

The information in this block should be exactly as approved by CUHK Senate. In case there are any differences, please explain in the table below.

Course code	MDS5202
Course title (English)	Applied Regression Analysis
Course title (Chinese)	应用回归分析
Units	3
Description (English)	This course introduces the basic theory and methodologies of regression analysis and demonstrates its practical applications. Topics include simple linear regression, multiple regression, generalized linear models, logistic regression, log-linear models, high-dimensional regressions, EM algorithm for regressions.
Description (Chinese)	本课程介绍回归分析的基本理论与方法，并展示其实际应用。内容包括：简单线性回归、多元回归、广义线性模型、逻辑回归、对数线性模型、高维回归，以及用于回归分析的EM算法。

B. Corresponding course at CUHK

Please give details of the closest corresponding course at CUHK (as approved by CUHK Senate and listed in course list). If the course in Shenzhen maps to more than one course at CUHK, please make multiple copies of the block below.

Course code	
Course title (English)	
Course title (Chinese)	
Units	
Description (English)	
Description (Chinese)	

2. Course requisites

Please state prerequisites and co-requisites, in terms of courses at CUHK-Shenzhen or any other requirements.

A. Prerequisites

B. Co-requisites

C. Exclusion

D. Others

3. Learning outcomes

1. Students should be able to analyze datasets using appropriate methods.
2. Students should be able to derive parameter estimation, construct hypothesis testing for different models.
3. Students should be able to assess model performance and make comparisons between models.
4. Student should learn the practical use of the R statistical package when applied to data analyses.

4. Course syllabus

Please highlight fundamental concepts involved in each topic to help students better understand what is covered in the course.

1. Simple linear regression
2. Multiple linear regression
3. Generalized linear models
4. Logistic regressions and log-linear models
5. High-dimensional regressions
6. EM algorithm and applications in regression

5. Assessment scheme

Component/ method	% weight
Assignment	60
Final exam	40

Remarks:

6. Grade descriptor

The grade descriptor should align with specific course learning outcomes, not a generic description applicable to all courses.

Type:**General**

Grade	Description
A	Demonstrate thorough mastery at an advanced level of extensive knowledge and skills required for attaining all the course learning outcomes. Show strong analytical and critical abilities and logical thinking, with evidence of original thought, and ability to apply knowledge to a wide range of complex, familiar and unfamiliar situations. Apply highly effective programming, organizational and presentational skills.
B	Demonstrate substantial command of a broad range of knowledge and skills required for attaining at least most of the course learning outcomes. Show evidence of analytical and critical abilities and logical thinking, and ability to apply knowledge to familiar and some unfamiliar situations. Apply effective programming, organizational and presentational skills.
C	Demonstrate general but incomplete command of knowledge and skills required for attaining most of the course learning outcomes. Show evidence of some analytical and critical abilities and logical thinking, and ability to apply knowledge to most familiar situations. Apply moderately effective programming, organizational and presentational skills.
D	Demonstrate partial but limited command of knowledge and skills required for attaining some of the course learning outcomes. Show evidence of some coherent and logical thinking, but with limited analytical and critical abilities. Show limited ability to apply knowledge to solve problems. Apply limited or barely effective programming, organizational and presentational skills.
F	Demonstrate little or no evidence of command of knowledge and skills required for attaining the course learning outcomes. Lack of analytical and critical abilities, logical and coherent thinking. Show very little or no ability to apply knowledge to solve problems. Programming, organizational and presentational skills are minimally effective or ineffective.

7. Feedback for evaluation

Please specify forms of evaluation to generate feedback on classes from students.

Formal Course and Teaching Evaluation
Informal feedback and/or teaching assistant

8. Readings

Required:

1.The Elements of Statistical Learning, 978-0-387-84857-0, Trevor Hastie, Robert Tibshirani, and Jerome Friedman, 1ST, United States

2.Categorical Data Analysis, 9780470463635, Alan Agresti, 3RD, United States

3.Introduction to regression modeling, 2-7, 9780534420758, Abraham and Ledolter, 1ST, United States

Others:

9. Course components

Activity	Hours/week
Lectures	3

10. Indicative teaching plan

Week	Content/ topic/ activity
1	Simple linear regressions
2	Simple/Multiple linear regressions
3	Multiple linear regressions
4	Multiple linear regressions
5	Model diagnostic
6	Generalized linear models
7	Generalized linear models
8	Generalized linear models
9	Logistic regressions
10	Log-linear models
11	EM algorithms
12	High-dimensional regressions
13	High-dimensional regressions
14	Review

11. Implementation plan

The implementation plan may vary from year to year. Please indicate expected enrolment, and number of sections. Example: 150 students for lecture (x 2); 30 students for tutorials (x 10).

12. Academic honesty and plagiarism

A course outline may include subject-specific requirements on plagiarism, in addition to the University established policies.

Students must complete all coursework and assessments independently, unless AI tool use is explicitly permitted. Unauthorized or improper use of AI may constitute academic dishonesty and harm learning. Misuse includes submitting AI-generated work as one's own, using AI to cheat, or employing AI unethically. Always obtain permission before using AI in academic work, and be sure to acknowledge any AI use in submissions.

13. Approval

Has the course title been included in the programme submission approved by CUHK Senate? Are there any differences? Have the details (as in this document) been approved at School or other level in CUHK- Shenzhen?

退回修改。

14. Any other information

15. Version date

Version number	1
As of (date)	2025-10-11